AP PRECALCULUS

Unit 1 • Worksheet 1.5A • Answer key

Worksheet 1.5A • Polynomial Functions and Complex Zeros

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What are the zeros of $p(x) = x^2(x-4)(x+1)$?

(A) $0,\ 4,\ -1$

(B) $0,\ -4,\ 1$

(C) $4,\ -1$

(D) $2,\ 4,\ -1$

Solution

Answer: (A). Set each factor to 0.

Question 2

NO CALCULATOR

$p(3) = 0$ for a polynomial p. Which must be true?

(A) $(x+3)$ is a factor of $p(x)$.

(B) $(x-3)$ is a factor of $p(x)$.

(C) $p(0) = 3$.

(D) p has degree 3.

Solution

Answer: (B). A zero a corresponds to the factor $(x-a)$.

Question 3

NO CALCULATOR

Which polynomial has x-intercepts at -2, 1 and 5 only?

(A) $(x-2)(x+1)(x+5)$

(B) $(x+2)(x-1)(x-5)$

(C) $(x+2)(x-1)(x+5)$

(D) $x(x+2)(x-1)(x-5)$

Solution

Answer: (B). The factors $(x+2)$, $(x-1)$, $(x-5)$ give zeros $-2, 1, 5$. Choice (D) also has the intercept 0.

Question 4 (Free response) NO CALCULATOR

Find all real zeros of each polynomial.

(a) $f(x) = x^3 - 9x$ (b) $g(x) = x^3 - 2x^2 - 8x$
(c) $h(x) = x^4 - 13x^2 + 36$ (d) $k(x) = 2x^2 - x - 3$

Solution
(a) $x(x-3)(x+3)$: $0, 3, -3$. (b) $x(x-4)(x+2)$: $0, 4, -2$.
(c) $(x^2-4)(x^2-9)$: $\pm 2, \pm 3$. (d) $(2x-3)(x+1)$: $\frac{3}{2}, -1$.